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The Effect of Operational Process Changes on Preoperative Patient Flow: Evidence from Field Research

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We partnered with a leading US academic medical center to empirically examine the impact of operational process changes designed to improve preoperative flow of patients through the perioperative environment. We focus on the implementation of centralized decision-making and the introduction of an information technology (IT)-enabled intraoperative prompt, on efficiency, as measured by preoperative patient processing time. We analyze over 33,000 individual surgical cases in a unique field experimental setting to conduct an empirical investigation of the effects of each intervention on patients' time spent in preoperative processing. To identify the *causal* effect of each process change, we leverage our field experimental research design and cast our analyses in the difference-in-differences modeling framework. We compare the preoperative patient processing time of two distinct patient groups, a *treatment group* that is impacted by the implemented operational changes and a *control group* that was not impacted by the changes, *before* and *after* each process change. Our results suggest a 3.4% reduction in preoperative processing time with only centralized decision-making in place, yet suggest a 10.8% reduction in preoperative processing time when centralized decision-making is paired with the IT-enabled intraoperative prompt. We also find evidence of a complementarity effect between our process changes and surgeon prior process experience. Our study contributes to the healthcare operations literature by demonstrating the benefits of coordinated information flow within the patient supply chain. We offer insights for hospital managers and healthcare operations scholars alike on the role of information coordination in improving preoperative patient flow with minimal impact on existing resources and staff.

Key words: centralized decision-making; preoperative; patient flow; healthcare operations; difference-in-differences
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1. Introduction

The Institute for Healthcare Optimization (2017) estimates that proper management of patient flow can reduce healthcare costs in the United States by 4–5%, or $35 to $112 billion, annually. In no area of hospital operations is this more important than the perioperative environment. Research suggests that surgical procedures are performed during 28% of hospital stays in the United States, with those hospital stays accounting for nearly half (49.1%) of all hospital costs (Fingar et al. 2014). With hospitals incurring costs of $36 or more per minute of operating room time, managing process flow in the perioperative services department is critical to improving process efficiency and cost performance (Childers and Maggard-Gibbons 2018).

Prior healthcare operations literature has established patient flow as a primary source of operational inefficiency within the hospital, with the goal of improving outcomes such as reductions in cost, improved operating efficiency, and enhanced patient safety (e.g., Christian et al. 2006, Fredendall et al. 2009, Friedman et al. 2006, Stahl et al. 2006). While there has been much attention in the healthcare operations literature afforded to patient flow from the perspective of queuing theory (e.g., Best et al. 2015, Deo and Gurvich 2011, Song et al., 2015b), fewer studies have leveraged lessons learned in supply chain management and its application to patient flow. Our study addresses this gap in the healthcare operations literature by building upon theory grounded in the supply chain management literature and integrating it into the unique context present in the healthcare setting. In doing so, we empirically demonstrate an alternative, and unexplored, path to achieve efficiencies in patient flow.

One reason why traditional product supply chain concepts seldom travel to service supply chains, and the healthcare setting in particular, is the belief that healthcare processes are fundamentally different from all other service processes, especially product manufacturing processes, where producing large quantities of highly similar products often leads to "standardized and centralized procedures" (Sengupta et al. 2006). In fact, prior research examining healthcare process improvement asserts that a principal challenge of applying standardized and centralized processes is the reality that patients have different needs (Drotz and Poksinska 2014, Timmermans and Berg 2003). Building upon the process view of supply chain management (Croxton et al. 2001), our study demonstrates how the coordination of activities and information flow within the patient supply chain, such as the perioperative services department of a hospital, presents a distinct opportunity for improvements in patient flow, despite the presence of high levels of variance and uncertainty associated with patients as process inputs.

Hospital perioperative services are generally composed of three distinct phases, the preoperative phase, intraoperative phase, and postoperative phase. Existing literature in the area of perioperative process improvement has predominantly focused on intraoperative (Fong et al. 2016) and postoperative (Adams et al. 2004) changes to improve patient flow through perioperative services. To the best of our knowledge, no study has examined the effect of operational interventions specifically designed to improve preoperative patient flow, an area of opportunity which often constitutes a significant source of process delays and is known to impact patients' perceptions of care (Gilmartin and Wright 2008, Tiwari et al. 2017). Addressing this gap in the healthcare operations literature, we conduct a field experiment to investigate the effect of process and technological interventions, which have proven effective in traditional supply chain settings, on preoperative patient flow.

Operational process changes are traditionally classified into two categories: process redesign and information technology (IT)-enabled changes (Davenport and Short 1990). Accordingly, we aim to improve preoperative patient flow through a focus on two distinct, yet interdependent, operational changes implemented within the perioperative environment of our focal hospital: a process redesign to centralize decision-making within the perioperative space, followed by an intraoperative IT prompt to enhance the coordination of information shared within the department. Centralized decision-making has proven effective in improving the flow of goods in supply chains that generate value through the movement of highly similar products (Clark and Scarf 1960, Sengupta et al. 2006), yet less evidence exists on the application of this concept to complex service supply chains, such as patient flow. In an effort to resolve this gap in the literature, we examine whether the performance benefits of centralized decision-making in product manufacturing supply chains translate to the services setting, and the perioperative services context in particular. Thus, our *first* objective is to examine the impact of an operational process change from decentralized to centralized decision-making on the preoperative flow of patients through the perioperative environment.

Access to real-time information across various decision-making units can also lead to improvements in operational decision-making (Stalk and Paranikas 2015). Coordinated integration of information provides suppliers with more accurate downstream data, which can be incorporated into operational decision-making, ultimately leading to reductions in inefficiencies in the flow of materials throughout the supply chain. Although prior studies in healthcare operations have examined how hospitals augment their operational practices with information technology (IT) with the goal of improving quality, lowering error rates, and increasing revenues (e.g., Aron et al. 2011, Demirezen et al. 2016, Guha and Kumar 2018, Janakiraman et al. 2017, Rajapakshe et al. 2020), a limited set of studies has examined the role of IT in improving patient flow. Further, no study, to the best of our knowledge, has specifically examined the causal impact of IT-enabled change on preoperative patient flow, which can yield substantial efficiency improvements to patient flow throughout the perioperative environment. Thus, our *second* objective is to examine the effect of a technological change that facilitates coordinated information sharing on preoperative patient flow within the perioperative environment.

To accomplish our research objectives, we partnered with a leading US academic medical center to implement two operational process changes, centralized decision-making followed by the introduction of an IT-enabled intraoperative prompt. A novel feature of our field data is the temporal separation of the two process changes within our focal hospital; more specifically, the intraoperative IT prompt was introduced after the implementation of centralized decision-making. This temporal separation enables us to empirically investigate the incremental benefit to preoperative patient flow from the introduction of the intraoperative IT prompt, following successful implementation of the operational change to centralize decision-making. However, a simple comparison of average patient flow before and after each process change is not sufficient to rule out confounding factors which may be unobservable to researchers. To establish the causal effect of the proposed changes on preoperative patient flow, we employ the difference-in-differences (DID) research design (Card and Krueger 1993) that is popular in the program evaluation literature and that has been recently applied in the context of healthcare operations (Song et al., 2015a,b, Song et al. 2018). Our DID model compares the preoperative length of stay of two distinct types of surgical cases—a control group of cases (the first case of each day in each operating room) that were *not* impacted by the introduction of our operational process changes and a treatment group of cases (the non-first cases of each day in each operating room) that were impacted by the serial introduction of our operational process changes—*before* and *after* the process changes. Our DID model also accounts for physician specific fixed effects, patient classification, and a rich set of control variables that account for temporal factors in the form of month of the year, day of the week, and time of day.

We find that the operational process change to centralize decision-making yields a 3.4% improvement in preoperative patient flow. Our results further indicate that the implementation of an intraoperative IT prompt aimed at increasing the coordination of activities and information flow results in a 10.8% reduction in preoperative length of stay (inclusive of the benefits from centralized decision-making). Further, we find evidence of the moderating effect of surgeon prior process experience on the relationship between the process changes and preoperative patient flow, such that surgeons with pre-implementation process experience generated greater reductions to preoperative patient flow than their colleagues who were previously conducting procedures at an alternative location.

Our study contributes to the healthcare operations literature by providing empirical support for the integration of supply chain process principles into the highly complex healthcare service setting. We further contribute to the healthcare operations literature, particularly the discussion surrounding the relationship between IT and operations, through our empirical examination of IT process changes which support and enhance the benefits of a previously implemented operational process change. We also demonstrate evidence of a complementarity effect between prior process experience of surgeons and the effectiveness of these operational changes on preoperative patient flow. In doing so, we contribute to practice by providing hospital managers and surgeons with empirical evidence of operational process changes which increase the coordination of activities and information flow within the perioperative environment. As a result, our findings have implications for both efficiency improvements to patient flow and significant cost savings to the hospital. Our study also contributes to the literature by highlighting the role of centralized decision-making and IT-enabled information sharing mechanisms in knowledge-intensive service operations. Last, our research joins a limited set of studies in operations management that employ a field experimental research design to establish the causal impact of the focal variable of interest.

The remainder of the study is organized as follows. A review of the theoretical background and development of our empirical propositions is outlined in section 2. We define the problem facing our focal hospital along with a detailed description of the research setting and dataset that we leverage to solve our research questions in section 3. We state the models employed and the analytical methods selected to test our propositions in section 4. We report results in section 5, followed by alternative model specifications and robustness checks in section 6. Finally, we conclude with a discussion of findings and their implications for practice in section 7.

2. Research Background and Propositions

The Global Supply Chain Forum Framework (Croxton et al. 2001) relies upon processes and information that must be coordinated across organizations within a supply chain, and within the functions of an individual member of the supply chain, in order to improve performance across the overall supply chain. In fact, Croxton and colleagues (2001, p. 15) argue that the strategic management of processes and information flow within the firm is the "necessary first step in integrating the firm with other members of the supply chain," advocating for a foundational level of intra-firm, cross-functional coordination of activities and information flow. In the services supply chain literature, and relevant to our setting, Aronsson et al. (2011) propose a conceptual framework which draws parallels between the movement of goods through a supply chain and the flow of patients through a hospital, such that various medical functions (i.e., departments) within a hospital represent individual firms in a patient supply chain. Viewing patient flow through this lens, Aronsson and colleagues (2011) conclude that the deployment of supply chain management practices in the hospital setting represents an opportunity for generating improvements to patient flow.

Integrating these two frameworks and building upon the process view of supply chains, we assert that intrafirm (i.e., intradepartmental) coordination of processes and information flow is required in complex service settings, particularly in the hospital environment where patients' lives depend upon accurate and seamless flow of activities and information across and within multiple units and departments. In other words, for an individual "firm" (such as a perioperative services department) in a large, complex patient supply chain (such as a hospital) to adequately coordinate processes and information with other firms (medical functions) across the supply chain (hospital), a foundational level of intrafirm (intradepartmental) coordination must first be achieved. Only then will the supply chain (hospital) be sufficiently prepared to realize the performance benefits from interfirm (interdepartmental) coordination of activities and information flow across the breadth of the patient supply chain. In this study, we build upon the process view of supply chains and leverage the parallels between the flow of goods in traditional supply chains and the flow of patients within a hospital supply chain. To do so, we employ a field experimental research design to test the application of supply chain principles in the hospital setting, addressing a gap in the healthcare operations literature. The supply chain principles deployed in our study setting, centralized decision-making and an intraoperative IT prompt, were specifically designed to enhance the coordination of processes and information at the intrafirm (intradepartmental) level of the patient supply chain, with the goal of improving preoperative patient flow.

2.1. Centralized Decision-Making

Patient flow through hospitals is a highly complex logistical undertaking, often marked by a lack of ownership and poor coordination between departments (Haraden and Resar 2004, Villa et al. 2009). As such, healthcare operations scholars have examined the impact of process improvement initiatives on patient flow. Extant studies have examined issues related to improving efficiency within perioperative services, whether through a focus on surgical scheduling (May et al. 2011), aggregate departmental processes (Cima et al. 2011, Fredendall et al. 2009), or postoperative turnaround time (Adams et al. 2004, Harders et al. 2006), albeit without a distinct focus on preoperative patient flow. We contextually distinguish our study from prior examinations of perioperative process improvement by focusing on the impact of changes specifically designed to reduce inefficiencies in patient flow in the preoperative phase of a perioperative services department.

Prior to our engagement with our study hospital, preoperative processing in the focal hospital could be initiated by any number of staff. This decentralized decision-making system resulted in highly variable preoperative patient arrival, excessive waiting time, and process bottlenecks. Seminal work in inventory management has provided evidence that centralized decision-making can optimize performance outcomes (Clark and Scarf 1960). Further, prior review of the coordination of information flow and materials through product supply chains has emphasized the process benefits that can be achieved through centralized decision-making (Sahin and Robinson 2002). Namely, supply chains are optimized when a central decision-maker acts with the best interest of the overall system in mind. While much evidence supporting this belief has been conducted in the traditional product manufacturing environment, decidedly less attention has been afforded to the application of centralized decision-making in service settings, particularly within complex environments such as those found in hospitals.

We address this gap in the healthcare operations literature through the application of centralized decision-making to coordinate the flow of patients within the perioperative services department. We argue that the presence of a centralized decision-maker, with a holistic view of patient flow through perioperative services, will reduce the inefficiencies present in a highly complex system with multiple decentralized decision-makers. In doing so, we aim to establish evidence of the benefits to preoperative patient flow from the coordination of processes and information at the intrafirm (e.g., intradepartmental) level of the patient supply chain, in spite of the challenges presented by heterogeneous patient inputs. Based on the above arguments, we incorporate the patient flow process improvement literature with concepts from the field of supply chain management and offer the following proposition:

Proposition 1. *An operational process change to centralize decision-making within the perioperative environment will result in a reduction in preoperative processing time.*

2.2. Intraoperative IT Prompt

Prior investigation of the impact of information sharing in product-based supply chains has provided evidence of cost and inventory reductions for the manufacturer (Gavirneni et al. 1999, Lee et al. 2000) due to enhancements in the visibility of materials flow throughout the chain (Cachon and Fisher 2000, Cao et al. 2014, Christopher 2000). More closely related to our setting, studies in the healthcare operations literature have illustrated the benefits of the presence and utilization of IT in hospital settings on performance outcomes, such as reductions in medical errors and improved financial performance (e.g., Aron et al. 2011, Devaraj and Kohli 2003, Devaraj et al. 2013). A prevailing theme within this stream of literature is that IT supports the coordination of information sharing within and across the organization, while also reducing opportunities for the introduction of process variation resulting from human behavior. Specifically, IT can be implemented to routinize process change by reminding users of standard processes and reducing the barriers for following such processes. As an example, computerized physician order entry systems have been implemented to adapt physician ordering behaviors while electronic monitoring has been employed to encourage users to wash hands, both of which reduce the barriers to comply with clinical best practices (Dai et al. 2015, Davidson and Chismar 2007). Leveraging these successes in other areas of hospital operations, we examine how the IT systems present in the perioperative department can improve the accuracy and coordinated flow of information to centralized decision-makers within the patient supply chain.

Enhancements to information sharing within the perioperative environment can be achieved through implementation of an intraoperative prompt. An intraoperative prompt is an IT solution designed to elicit real-time information (from within the operating room) with respect to the progress of an ongoing surgical case. As an example of its application in the surgical setting, Dexter et al. (2009) outline the implementation of an intraoperative IT prompt through a negotiated interruption that requests an update on surgery end time for cases that are running late. Building upon the intraoperative prompt implemented by Dexter et al. (2009), the IT prompt implemented in our setting represents a scheduled interruption that occurs at a pre-specified time (60 minutes prior to the previously scheduled end time) and applies universally to all cases.[1]

While much of the existing research on information sharing in hospitals focuses overwhelmingly on the role that shared information plays in providing multidisciplinary clinical care to patients (Fichman et al. 2011, Walker et al. 2005), our study advocates for an alternative benefit of information sharing in the clinical setting. Through the application of lessons learned in information systems, supply chain management and behavioral research we propose employing coordinated information to aid the flow of patients within a focused area of the hospital setting. As such, we test whether the availability of the intraoperative prompt leads to the enhanced coordination and accuracy of information flow from the operating room to the centralized decision-maker, leading to a reduction in preoperative processing time. We thus offer the following proposition:

Proposition 2. *The introduction of an intraoperative IT prompt will result in a further reduction in preoperative processing time, relative to the previously implemented operational process change to centralize decision-making within the perioperative environment.*

2.3. Surgeon Prior Process Experience

The extent to which benefits of the operational process change are realized may vary across the entities or individuals involved in service delivery. Within our context, surgeons play a critical role in initiating the flow of intraoperative information (about the current case) to the centralized decision-maker responsible for initiating the preoperative process for the next scheduled patient (in the same operating room location). Prior studies in the medical literature have also taken an interest in examining the role of individual surgeon characteristics, highlighting the impact of clinicians' prior experience on both clinical and economic outcomes (Meltzer et al. 2002, Neumayer et al. 2005, Sosa et al. 1998). Building on these studies, and since the surgeons in our setting possess considerable influence over an elementary component of preoperative flow, we propose that *surgeon-specific* experience with the on-site process will moderate the benefits of our process change interventions. In other words, the improvements to preoperative patient flow resulting from the implementation of our process changes will be differentially affected by the heterogeneous process experience of the surgeons in our setting.

Complementarity theory suggests that two activities are complements of each other if "doing more of one thing increases the returns to doing more of another" (Milgrom and Roberts 1995, p. 181). From an organizational perspective, two resources are complementary if having one increases the returns of having another resource. Several studies in the information systems and management science literature have relied on this complementarity perspective to understand how the interaction between information technology and other organizational resources leads to improvements in organizational performance (e.g., Barua et al. 1996, Bharadwaj et al. 2007, Brynjolfsson and Hitt 1998). Bharadwaj et al. (2007) argue that IT capabilities enhance the coordination processes of an organization and therefore will amplify the positive effects of manufacturer–marketing and manufacturer–supply chain coordination on manufacturing performance. More closely related to our hospital context, prior work has applied complementarity theory to explain the impact of the relationship between quality management training and IT automation on a reduction in medical error rates of hospitals (Aron et al. 2011).

Reagans et al. (2005), in their analysis of how organizations benefit from various types of experience, note the role of experience as a key influencer of task performance, such that experience is believed to lead to more effective coordination of information and behavior between distinct roles. In the context of healthcare, Janakiraman et al. (2017) document that physicians with greater experience with a health information exchange are able to better leverage the information that is available through the technology platform. Within our setting, we believe that surgeons with greater prior on-site experience will have higher levels of process experience, which may enhance the accuracy of information shared with the centralized decision-maker. As such, we hypothesize that this process familiarity will moderate the effectiveness of the relationship between coordinated activities and information flow and improvements to preoperative patient flow.

Alternatively, we acknowledge that a counterargument exists such that surgeons with greater pre-implementation process experience may be reluctant to change their behavior to comply with new operational processes. This is particularly true since a surgeon's response to the intraoperative IT prompt benefits the surgeon on the next scheduled case, thus there are no immediate process benefits for the surgeon on the existing case. However, we believe that surgeons with prior process experience will be further motivated by their negative experiences with the pre-intervention process and will thus act in the best interest of overall departmental efficiency.

Leveraging complementarity theory, we propose that surgeons who have more experience with the pre-implementation process will generate more effective and accurate information sharing, leading to greater reductions in preoperative processing, than their peers who have little experience with the pre-implementation process. In other words, a surgeon's prior process experience will enhance the preoperative benefits realized through the implementation of IT-enabled centralized decision-making, leading to the following proposition:

PROPOSITION 3. *Surgeon prior process experience will amplify the relationship between the previously implemented operational process changes and preoperative processing time.*

3. Data

The data are derived from a field study conducted in an academic medical center in the United States that completes over 28,000 prescheduled (non-emergency) surgeries annually in 39 operating rooms serviced by 28 surgical specialties. One of the authors, in collaboration with surgeons, anesthesiologists, and department administrators at our focal hospital, was directly responsible for the design and implementation of the process changes outlined in this section. The surgical process studied here begins with a request to the holding room staff (by the intraoperative surgical staff) to bring the next scheduled surgical patient to the preoperative holding room. Once in the holding room, the patient undergoes final preparation for surgery before transport to the operating room for anesthesia induction and surgery. Upon completion of surgery, the patient travels to the post-anesthesia care unit for initial recovery, and then transitions to their ultimate destination dependent upon the patient's status classification. With respect to the overall flow of patients through the perioperative environment, this study focuses on operational improvements targeted at improving the holding room processing time, that is, the time between the arrival of a patient to the preoperative holding room and the entry of the patient into the operating room.

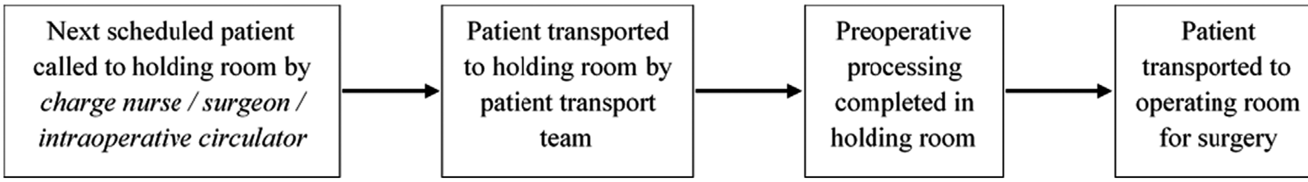
3.1. Pre-Intervention

Prior to the process improvement intervention to centralize decision-making, any number of individuals could call for the next scheduled patient to be transported to the holding room, a system we refer to as decentralized decision-making. As a result, the surgical department frequently experienced high patient processing times in the holding room. This phenomenon was exacerbated by the fact that there are 39 operating rooms being supplied by 23 holding rooms. The immediate ramifications of this unsynchronized flow include a suboptimal process due to unpredictability in the arrival of inputs (patients), excessive waiting time, and process bottlenecks. A potential longer term consequence facing our study hospital was the need to build additional holding room space, an extremely costly and highly difficult logistical undertaking. This process is depicted in Figure 1.

3.2. Centralized Decision-Making

To reduce the time that patients spend in the holding room prior to surgery, we relied on lessons learned in product-based supply chain settings and applied the concept of centralized decision-making to the perioperative arena. As outlined in section 2.1, we implemented an operational process change to centralize

Figure 1 Pre-Intervention Process Flow (Decentralized Decision-making)



the decision-making responsibility for calling the next patient to the holding room. The operational process change enacted narrowed the responsibility for calling the next patient to the holding room by assigning the task to the holding room charge nurse only, with the goal of reducing processing time in the holding room. The holding room charge nurse is stationed at a central location within the perioperative department and has access to a surgery and patient tracking system that provides a holistic view of department operations (analogous to the role of air traffic controllers at an airport). The tracking system provides a visual display of timestamps relevant to the current surgical case in each operating room, enabling the centralized decision-maker to assess the progress of each surgery and call for the next patient assigned to a specific operating room when they believe that the current surgery is approximately 60 minutes from completion, a time point selected through input from anesthesiologists and surgeons in our study setting. This process is depicted in Figure 2.

3.3. Intraoperative IT Prompt

Approximately six months later, we implemented an intraoperative IT prompt. The prompt represented in our model is an automatic electronic prompt that appears via a "pop-up" window within the intraoperative nursing screen (for the first time) 60 minutes prior to the originally scheduled end time of the surgery in progress.[2] The operating room circulator is responsible for responding to the prompt, following consultation with the surgeon about the expected end time of the case. If the surgeon responds to the initial prompt that more time is needed to complete the surgery, the circulator will revise the originally scheduled end time to reflect an updated surgery end time. Assuming the surgery will now take longer than the originally scheduled end time (which is most often the case), the new end time is greater than 60 minutes from the time in which it was modified in the intraoperative documentation. The intraoperative prompt will then reappear 60 minutes prior to the revised end time for the surgery in progress. In other words, the intraoperative prompt will continue to appear as long as the surgery end time continues to be delayed from its originally intended completion time. In the unlikely, but possible, event that the surgery will finish sooner than the originally scheduled end time, following receipt of the initial intraoperative prompt the operating room circulator can also update the scheduled end time to reflect a quicker finish. Such an occurrence would immediately trigger the holding room charge nurse to call the next scheduled patient (for that operating room location) to the holding room to begin preoperative processing. We note that our decision to implement the intraoperative IT prompt after centralized decision-making was motivated both by prior literature (Aron et al. 2011) and the practical nature of our study setting, which necessitates that a centralized decision-maker be established in order to act upon information from the intraoperative IT prompt. This process is depicted in Figure 3.

3.4. Implementation Timeline

Accurately assessing the effectiveness of process changes implemented within an organization necessitates consideration of other factors which may influence intended process outcomes. Although we empirically address potential confounders through our selected empirical methodology, we also plan for potential confounders through a careful survey of our focal hospital's organizational environment prior to selecting our implementation timeline. Organizational change literature has previously established

Figure 2 Centralized Decision-making Process Flow

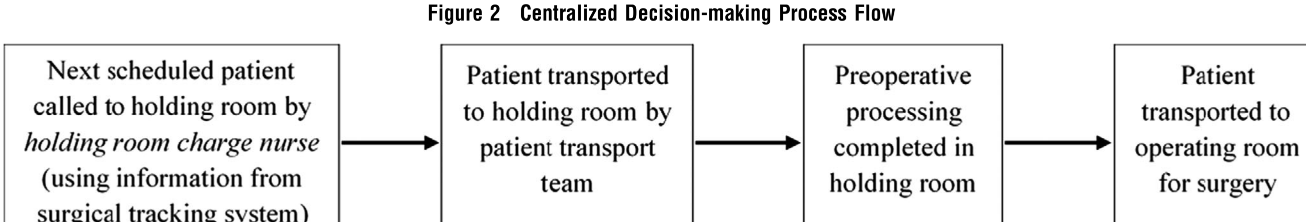
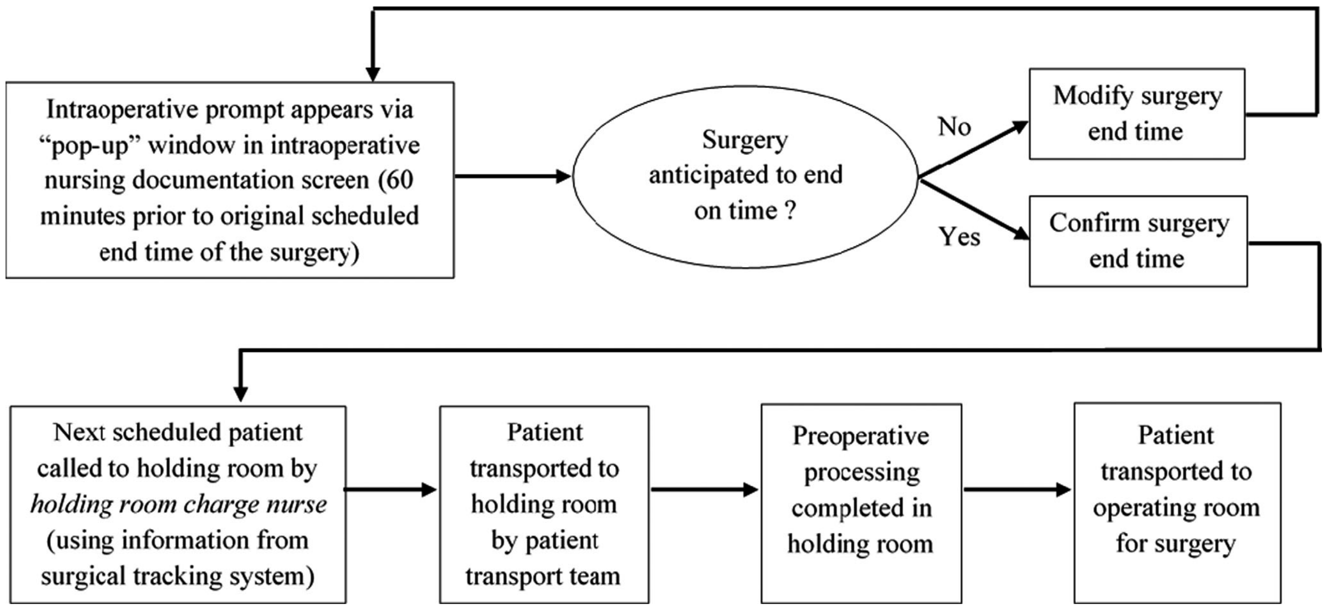


Figure 3 Intraoperative IT Prompt Process Flow



that the timing of change initiatives is often dependent upon the organization's environment, both internal and external (Hinings and Greenwood 1989). Within our setting, each process change was deliberately planned to occur during a period of time in which no other major process or staffing initiatives were planned to take effect. To illustrate this point, we examine the timing of each process change launch with respect to a major annual staffing initiative which has been empirically demonstrated to have a significant impact on surgical department processes and performance, the simultaneous exit and entry of surgical residents each July.

Each July in teaching hospitals across the United States, including that in which our study takes place, a planned exit of experienced (upper level) medical residents occurs jointly with the planned arrival of a similar number of inexperienced (first year) medical residents, a phenomenon which Song et al. (2015a) deemed as cohort turnover. The Medicare Payment Advisory Commission (MedPAC) defines a major teaching hospital as that with a teaching intensity (resident to licensed bed ratio) of 0.25 or greater (Medicare Payment Advisory Commission 2002). The teaching intensity in our study hospital is above 0.85, indicative of a highly teaching intensive setting. Song et al. (2015a) empirically demonstrated that hospitals with high levels of teaching intensity (like our focal hospital) often experience a negative impact to operational performance in the months immediately following this "July phenomenon," particularly in the months of July and August. Mindful that the annual cohort turnover in our setting would more than likely negatively impact operational performance and potentially introduce a confounding effect in our study setting, we carefully selected the months in which our process changes were introduced, April and November, to avoid the "July phenomenon."

Additionally, we deliberately planned the process change launch schedule to avoid changes occurring immediately before or after major holiday periods when large numbers of staff and surgeons are working reduced or modified schedules and conducting fewer prescheduled elective cases. Although the intraoperative prompt was implemented at the beginning of November, the planning team believed that this launch timing was far enough in advance of the end of year holidays. The design and implementation of our process changes were thoughtfully planned to ensure that they occurred during times of staff stability, or at least to avoid expected or planned staff instability. Lastly, recognizing the need for the patient flow process to reach a "steady state" before further changes were enacted, we coordinated the introduction of the intraoperative IT prompt to occur after a relative level of process stability had been reached following the process change to centralized decision-making.

3.5. Estimation Dataset

We began our analysis with 35,527 unique surgical cases in the dataset but excluded surgical cases due to a variety of reasons. We removed 2199 cases from our dataset due to incomplete information, primarily due to lack of a recorded timestamp for a patient entering the holding room.[3] We also excluded 232 cases due to negative elapsed time in the operating room and 39 cases due to negative elapsed time in the holding room, indicating an incorrect capture of patient flow through the surgical department, resulting in a final available sample of 33,057 surgical cases.

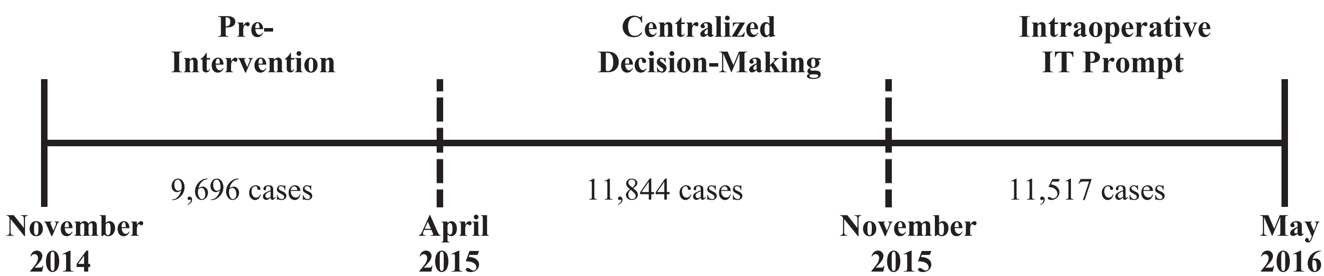
The final sample is composed of surgical cases belonging to three distinct phases of our study. We define the pre-intervention period of our study as the time prior to the implementation of centralized decision-making or the intraoperative IT prompt, which contains 9,696 cases and spans from November 3, 2014 to April 17, 2015 (approximately 5 ½ months). There are 11,844 cases in our final sample which occur after the implementation of centralized decision-making, but before the implementation of the intraoperative IT prompt, a period that spans from April 20, 2015 to October 30, 2015 (approximately 6 ½ months). Lastly, our sample contains 11,517 cases that occur between November 2, 2015 and May 13, 2016 (approximately 6 ½ months), after the implementation of the intraoperative IT prompt, which was employed in addition to the centralized decision-making already in practice. A summary timeline is displayed in Figure 4.

4. Identification Strategy

Our goal is to empirically analyze the impact of centralized decision-making and the intraoperative IT prompt on patients' time spent in the holding room. Examining the impact of the process changes at the departmental level may mask important surgical case patterns which cannot be detected at such aggregate levels; therefore, we conduct our analysis with individual patient-level surgical data from both before and after each implemented process change. A basic pre- and post-comparison, even with individual patient-level surgical process flow data, will not adequately rule out other environmental changes occurring during our study period that are unobservable to the researchers and may impact patient flow. To effectively eliminate the possibility of confounding factors impacting our dependent variable and to establish the causal impact of the two focal process changes, we invoke a natural experiment in which we examine the effect of the introduced process changes on patients' preoperative surgical flow, using the DID econometric approach. The DID approach allows for examination of the time spent in the holding room for surgical patients in a treatment group (those whose cases were impacted by the implemented process changes) and a control group (those patients whose cases were not impacted by the implemented process changes), before and after each implemented process improvement. In our setting, the DID econometric approach helps to establish the causal effects of each process change while also accounting for differences in the baseline holding room times across the two groups, as well as any unobservable changes impacting patient flow over time (Wooldridge 2002). The DID econometric specification has proven effective in analyzing the causal effects of policy and process adoption in recent operations and health policy analysis literature (Ayer et al. 2019, Gray et al. 2015, Khurana et al. 2019, Kumar et al. 2018, Rajaram et al. 2014, Schmitt 2017, Song et al. 2015a,b, Song et al. 2018).

The effectiveness of a DID approach is predicated upon the establishment of a theoretically and statistically robust control group, which is used to "difference" out any unobserved, endogenous factors which may influence a change in the outcome variable of interest. Recall that both the change to centralize decision-making and the intraoperative IT prompt required decision-makers to assess the progress of a current case before calling the next scheduled patient (for that same operating room) to the preoperative holding room. That is, each process change intervention will only have an impact on surgeries in which another surgical case is *immediately preceding* the case in question, in the *same* operating room location. As a result, the first scheduled case for each operating room on each day will not be affected by any preceding (or in progress) surgical case, establishing a natural control group against which we can compare the impact of our process changes. Thus, we define the control group as the set of first cases which occurred on the *same* day in the *same* operating room as the treatment group of surgical cases.

Figure 4 Study Period Timeline

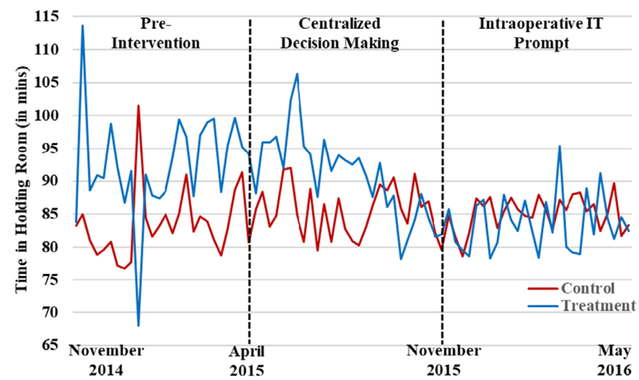


4.1. Empirical Investigation of Parallel Trend Assumption

A critical precursor for the use of the DID methodology requires that researchers demonstrate that there is no difference in trends between the treatment and control groups prior to the focal policy change. In other words, researchers must satisfy the parallel trend assumption which requires the trends in the outcome variable for the treatment and control group to be similar (or parallel) during the pre-intervention period. We begin our comparison of the treatment and control groups in our study by analyzing case volume within each time period of our study, across key surgical case grouping characteristics. Table 1 displays the percentages of surgical case-level covariates included in our analysis, segmented by time period (pre-intervention, centralized decision-making, and intraoperative IT prompt) and experimental (control vs. treatment) group. The percentage of surgical cases composing each covariate remains relatively consistent within the control and treatment groups across the three phases of our field experiment, providing evidence that our findings are not confounded by substantial changes in the volume composition of surgical cases over time.[4]

To substantiate that the parallel trend condition is satisfied in our data, we implemented a graphical analysis of the pre-intervention trends for patient time in the holding room for both the treatment and control groups within our dataset. Figure 5 shows a plot of the mean of patient time in the holding room prior to adjusting for time trends, fixed effects, or other

Figure 5 Comparison of Treatment Vs. Control Group Time in Holding Room [Color figure can be viewed at wileyonlinelibrary.com]



Notes. Data are collapsed at the weekly level for visual presentation. Dashed lines indicate implementation of centralized decision-making (April 2015) and the intraoperative IT prompt (November 2015).

covariates. The plot demonstrates strong evidence for the presence of parallel trends in the pre-intervention period of our study (November 2014 – April 2015). Further, mean patient time in the holding room drops significantly following the intervention in the treatment group, suggesting that our intervention resulted in a decrease in patient time in the holding room. To provide additional empirical support for parallel trends in the pre-intervention period, we estimate separate linear regression models with a continuous time variable to examine and compare the pre-intervention slopes for the control and treatment groups

Table 1 Surgical Case Distribution by Study Period and Experimental Group

	Control group			Treatment group		
	Pre-intervention	Centralized decision-making	Intraoperative IT prompt	Pre-intervention	Centralized decision-making	Intraoperative IT prompt
Day of week						
Monday	21.52%	19.43%	20.92%	21.67%	20.69%	21.38%
Tuesday	21.75%	21.30%	20.69%	22.66%	20.48%	19.72%
Wednesday	19.94%	20.49%	21.46%	18.60%	20.00%	21.31%
Thursday	18.69%	20.74%	19.96%	17.83%	21.12%	21.40%
Friday	18.09%	18.06%	16.98%	19.25%	17.71%	16.20%
Time of day						
Before noon	96.89%	96.65%	96.23%	47.17%	45.65%	44.19%
After noon	3.11%	3.35%	3.77%	52.83%	54.35%	55.81%
Surgery type						
Early morn admit	44.17%	44.71%	43.45%	26.59%	25.22%	24.23%
Inpatient	18.62%	19.14%	18.86%	27.98%	28.83%	30.15%
Observation	10.12%	11.72%	11.45%	9.64%	9.74%	9.43%
Same day surgery	27.09%	24.43%	26.24%	35.79%	36.21%	36.19%
Surgery duration						
Less than 1 hour	8.64%	8.81%	9.58%	18.12%	17.22%	17.67%
1–2 hours	22.50%	20.74%	20.77%	31.13%	32.71%	34.31%
2–3 hours	19.39%	19.68%	19.89%	21.81%	21.58%	21.19%
3–4 hours	15.41%	15.87%	17.53%	13.76%	13.28%	12.80%
> 4 hours	34.05%	34.91%	32.22%	15.18%	15.20%	14.03%
# Surgical cases	3991	4813	4665	5705	7031	6852

(Song et al. 2018). Post-estimation testing indicated no statistically significant difference ($p > 0.10$) in the pre-intervention slopes, supporting our assertion that the parallel paths assumption has been met. Taken together, these findings provide strong empirical support that there are *no observed differences between the treatment and control groups*, which in turn suggests that the parallel trend assumption holds in our context. We next proceed to analyzing the impact of our process changes on patients' time spent in the holding room.

4.2. Impact of Centralized Decision-Making

To test the effectiveness of the operational process change in Proposition 1, we propose the following DID model at the individual surgical case (i.e., patient) level:

$$\begin{aligned}\ln\left(TiHR_{ijt}\right) = \; & \beta_0 + \beta_1 Post_t + \beta_2 Trtmnt_{ij} \\ & + \beta_3\left(Post_t \times Trtmnt_{ij}\right) \\ & + \beta_4 ORTime_{ij} + Surgeon\_FE_j \\ & + PatientClass\_FE_i + Time\_FE_t \\ & + Location\_FE_i + \varepsilon_{ijt}.\end{aligned} \tag{1}$$

In this model, $\ln(TiHR_{ijt})$ denotes the log-transformed number of minutes that the patient corresponding to surgical case i spends in the holding room prior to a surgery conducted by surgeon j at time t; $Post_t$ equals 1 after the operational process change to centralize decision-making (but prior to the implementation of the intraoperative IT prompt) and 0 before the intervention; $Trtmnt_{ij}$ is a binary variable equal to 1 if surgical case i conducted by surgeon j belongs to the treatment group (non-first case of the day in any given operating room) and 0 if the surgical case belongs to the control group (first case of the day in any given operating room). With respect to Equation 1, the coefficient associated with the interaction term $Post_t \times Trtmnt_{ij}$ compares the difference in patients' time in the holding room before and after centralized decision-making for the treatment group vs. the control group. Proposition 1 predicts that the coefficient associated with this interaction term will be negative and significant, indicating that centralized decision-making resulted in a significant reduction in patients' time spent in the holding room.

The DID modeling framework that we employ helps compare surgical cases in the treatment group with surgical cases in the control group which occurred on the *same* day in the *same* room, eluding any endogeneity bias that may result from environmental factors (or omitted variables). Although discussion with process owners at the focal hospital confirms that no other significant operational changes occurred during our study period, we feel confident that any incremental changes which could have potentially affected our results would be seen equally in both the treatment and control groups.

We included several control variables in our analysis. In Equation 1, $ORTime_{ij}$ represents the number of minutes that the patient spends in the operating room (i.e., intraoperative time) for surgical case i conducted by surgeon j and serves as a control variable to capture the potential effects of surgical procedure length on preoperative processing time. Because our dataset is composed of prescheduled, elective cases, the surgeon on each case is scheduled *prior* to the patient's time in the holding room; therefore, we include $Surgeon\_FE_j$ as a block of control variables accounting for time invariant surgeon characteristics. $PatientClass\_FE_i$ is a block of control variables accounting for the surgical classification of patients. $Time\_FE_t$ includes the set of time of day, day of week, and month of year fixed effects. Finally, $Location\_FE_i$ includes dummy variables for each operating room location.

Within our study period, there are 2319 unique procedures completed, each represented by a distinct Current Procedural Terminology code (CPT). Many procedures are completed multiple times within our study period, leading to the potential for correlated model errors for patients undergoing the same procedure type. To avoid such within-procedure correlation, we choose to cluster standard errors by CPT as opposed to adhering to the traditional linear regression assumption that all standard errors are independently and identically distributed. Consideration was given to alternative clustering approaches, such as day or week level; however, time-based clustering is best reserved for panel data with the same individuals undergoing multiple procedures over time, thus enabling the model errors for the same individual in different time periods to be correlated (Cameron and Miller 2015). Since we do not have repeated patient observations (one surgery per patient in our dataset), clustering by procedure is the most appropriate empirical strategy for our dataset.

4.3. Impact of Intraoperative IT Prompt

We test the effect of the intraoperative IT prompt change in Proposition 2 by specifying the following DID model at the surgical case level:

$$\begin{aligned}\ln\left(TiHR_{ijt}\right) = \; & \beta_0 + \beta_1 Prompt_t + \beta_2 Trtmnt_{ij} \\ & + \beta_3\left(Prompt_t \times Trtmnt_{ij}\right) \\ & + \beta_4 ORTime_{ij} + Surgeon\_FE_j \\ & + PatientClass\_FE_i + Time\_FE_t \\ & + Location\_FE_i + \varepsilon_{ijt},\end{aligned} \tag{2}$$

In this model, $Prompt_t$ equals 1 after the intraoperative IT prompt change and 0 before the intervention; $Trtmnt_{ij}$ is a binary variable equal to 1 if surgical case i conducted by surgeon j belongs to the treatment group (non-first case of the day in any given operating room) and 0 if the surgical case belongs to the control group (first case of the day in any given operating room). All other control variables included in this model are identical to those in Equation 1.

In Equation 2, the coefficient associated with the interaction term $Prompt_t \times Trtmnt_{ij}$ is the DID estimator that lets us compare the difference in patients' time in the holding room before and after the intraoperative IT prompt change for the treatment group vs. the control group. Proposition 2 predicts that the coefficient on the interaction term will be negative and significant, indicating that patients spend less time in the holding room for non-first cases after the intraoperative IT prompt implementation.

4.4. Impact of Surgeon Prior Process Experience

To examine the effect of surgeon prior process experience, we leverage a novel aspect of our empirical context. The hospital under investigation in our study is the flagship hospital for a large regional healthcare network. The regional healthcare network, for which our study hospital is a part of, consists of hundreds of access locations, including multiple hospital and surgery center locations. Due to the healthcare network structure to which our study hospital belongs, prescheduled surgical procedures are routinely conducted at multiple care locations within the network. Embedded within the operational structure of completing a specific type of surgical procedure at alternative sites is that the specialized surgeons conducting those cases are practicing surgery at these alternative locations. This was indeed the case for prescheduled surgical procedures for two distinct surgical specialties within the healthcare network to which our study hospital belongs. Prior to the beginning of the pre-intervention phase in our study, the prescheduled surgical procedures for these two distinct surgical specialties were integrated into our study hospital for strategic reasons, whereas they were previously conducted at another location within the network. This decision to begin conducting new types of prescheduled surgical cases in our study hospital provides us with a clear demarcation of surgical procedure types (and surgeons) which are new to our focal hospital. As a result, the surgeons conducting these new types of surgical cases lack experience with the pre-intervention process at our site, as compared to surgeons who have been conducting their surgical cases at our study hospital for an extended period of time. Following this integration, we now had two types of surgeons in our data, those with extensive pre-intervention process experience and those new to our study site who lack experience with the pre-intervention processes.

To test the moderating impact of surgeon prior process experience on patients' length of stay in the holding room, we conducted an additional DID analysis by subdividing our treatment group into two subgroups. The first subgroup contained surgical cases assigned to the treatment group (non-first cases) and in which the surgeon on the immediately preceding case (in the same operating room location) had extensive pre-intervention process experience at the study site, a group denoted as *Treatment_Prior.* The second subgroup contained surgical cases assigned to the treatment group (non-first cases) and in which the surgeon on the immediately preceding case (in the same operating room location) did not have pre-intervention process experience at the study site, a group we refer to as *Treatment_NoPrior*. We test each of these subgroups in separate DID models while employing the same control variables as our primary analysis. Each of these models is presented in Equation 3 and 4, respectively.

$$\begin{aligned} \ln(TiHR_{ijt}) = {} & \beta_0 + \beta_1 Prompt_t + \beta_2 Trtmnt\_Prior_{ij} \\ & + \beta_3 (Prompt_t \times Trtmnt\_Prior_{ij}) \\ & + \beta_4 ORTime_{ij} + Surgeon\_FE_j \\ & + PatientClass\_FE_i + Time\_FE_t \\ & + Location\_FE_i + \varepsilon_{ijt}, \end{aligned} \tag{3}$$

$$\begin{aligned} \ln(TiHR_{ijt}) = {} & \beta_0 + \beta_1 Prompt_t + \beta_2 Trtmnt\_NoPrior_{ij} \\ & + \beta_3 (Prompt_t \times Trtmnt\_NoPrior_{ij}) \\ & + \beta_4 ORTime_{ij} + Surgeon\_FE_j \\ & + PatientClass\_FE_i + Time\_FE_t \\ & + Location\_FE_i + \varepsilon_{ijt}, \end{aligned} \tag{4}$$

5. Results

5.1. Impact of Centralized Decision-Making

The available set of patients included in the test for the effectiveness of centralized decision-making (Proposition 1) consists of the 9,696 surgical cases in the pre-intervention period and the 11,844 surgical cases following the implementation of the operational process change to centralize decision-making, but before the introduction of the intraoperative IT prompt. Table 2 displays the results of the DID estimation to test Proposition 1. The primary coefficient of interest is the one that corresponds to the interaction term $Post_t \times Trtmnt_{ij}$ and represents the percentage change in patients' time in the holding room as a

Table 2 Impact of Centralized Decision-making

	DV: ln(TiHR)
Post	−0.024 (0.0212)
Trtmnt	0.158*** (0.0138)
Post × Trtmnt	−0.034*** (0.0127)
OR Time (in mins)	3.09×10^{-04}*** (3.41×10^{-05})
Constant	4.509*** (0.0691)
Surgeon Fixed Effects	Yes
Patient Class Fixed Effects	Yes
Location Fixed Effects	Yes
Time Fixed Effects	Yes
Day Fixed Effects	Yes
Month Fixed Effects	Yes
Number of observations	21,539
R^2	0.201

Notes: Robust standard errors clustered by procedure type are in parentheses; $^{***}p < 0.01$.

result of a surgical case occurring after the operational process change for a non-first case in an operating room on a given day. We find that there is a significant reduction in patients' time spent ($\beta_3 = -0.034$; p-value = 0.007) in the holding room prior to surgery resulting from the operational process change to centralize decision-making. The non-first case patients who undergo a surgical procedure after centralized decision-making experience a 3.4% reduction in holding room length of stay (all other independent variables held constant), on average, supporting Proposition 1.

5.2. Impact of Intraoperative IT Prompt

The available set of patients included in the test for the effectiveness of the intraoperative IT prompt consists of the 9696 surgical cases in the pre-intervention period and the 11,517 surgical cases following the implementation of the intraoperative IT prompt. Table 3 displays the results of the DID estimation to test Proposition 2. The interaction term $Prompt_t \times Trtmnt_{ij}$ is the primary coefficient of interest which captures the reduction in treatment patients' holding room length of stay resulting from the implementation of the intraoperative IT prompt. We find that the intraoperative IT prompt results in a highly significant reduction ($\beta_3 = -0.108$; p-value = 0.000) in the holding room length of stay for patients in the treatment group, as predicted in Proposition 2. This result indicates that patients in the treatment group that undergo a surgical procedure after the implementation of the intraoperative IT prompt change experience a 10.8% reduction in holding room length of stay, on average (all other independent variables held constant). It is important to note that both changes were tested relative to the pre-intervention period of 9,696 available surgical cases. The results provide strong support for Proposition 2.

Table 3 Impact of Intraoperative IT Prompt

	DV: ln(TiHR)
Prompt	0.025*** (0.0087)
Trtmnt	0.150*** (0.0133)
Prompt × Trtmnt	−0.108*** (0.0137)
OR Time (in mins)	3.65×10^{-04}*** (3.83×10^{-05})
Constant	4.589*** (0.0800)
Surgeon Fixed Effects	Yes
Patient Class Fixed Effects	Yes
Location Fixed Effects	Yes
Time Fixed Effects	Yes
Day Fixed Effects	Yes
Month Fixed Effects	Yes
Number of observations	21,212
R^2	0.198

Notes: Robust standard errors clustered by procedure type are in parentheses; $^{***}p < 0.01$.

To validate this result, we conducted an alternative test comparing the 11,517 available surgical cases following the implementation of the intraoperative IT prompt with the 11,844 available surgical cases which occurred after the implementation of centralized decision-making (excluding the 9,696 available surgical cases in the pre-intervention period). Our results remain robust to the alternate specification, which we report in Table 5 in section 6.1.

5.3. Impact of Surgeon Prior Process Experience

With respect to testing the moderating effect of surgeon prior process experience, we considered several contextual factors before conducting the empirical test. Recall that the operational change to centralize decision-making places the responsibility to call the next scheduled patient to the holding room with the holding room charge nurse. The addition of the intraoperative IT prompt was designed to elicit a verification from the surgeon on an in progress case that the case is still projected to end at the previously scheduled end time. This confirmation (or modification) of the case end time is the definitive signal to the centralized decision-maker to begin the process of bringing the next scheduled patient (for that same operating room location) to the holding room. Because surgeons on in progress cases are initiating the information transfer (and the accuracy of the information being shared) to the centralized decision-maker, the effectiveness of the intraoperative prompt relies heavily on the judgment and process knowledge of the surgeon. As such, the surgeon's relative level of process experience will have a greater impact on the implementation of the intraoperative IT prompt than it will on the introduction of the operational process change to centralize decision-making. Additionally, recall from the test of Proposition 2 that the intraoperative prompt further reduces preoperative processing time (beyond the benefits already achieved by centralized decision-making). Since we have previously established that having both interventions in place generates the maximum benefit to preoperative patient flow, we believe it prudent to test the moderating impact of surgeon prior process experience on a time period when both process changes are in effect in our study setting. Thus, we select the period after the implementation of the intraoperative prompt as the appropriate period to test the moderating role of surgeon experience.

After selecting the appropriate time period within our sample for testing the impact of surgeon prior process experience, we now shift our focus to the operationalization of surgeon process experience. Recall that the surgeon on an existing (in progress) case provides information to the centralized decision-maker which will impact preoperative processing for the next scheduled case. Therefore, we must moderate the relationship between the intraoperative IT prompt and holding room length of stay with the prior process experience level of the surgeon on the *immediately preceding* case. In other words, it is the relative process experience of the surgeon on the preceding case (in the same operating room location) which is most relevant to studying the moderating role of prior process experience on the relationship between the intraoperative IT prompt and holding room length of stay.

Upon establishing the exact surgeon whose prior process experience is most influential to testing the moderating effect of prior process experience, we now turn to the most appropriate empirical method for testing this effect. We first consider implementing a difference-in-difference-in-differences (DDD) model which would involve a three-way interaction between the treatment (vs. control) dummy variable, the intraoperative prompt (vs. no prompt) dummy variable, and a third dummy variable which captures a surgeon's prior process experience (vs. surgeons lacking prior process experience). Unfortunately, this empirical strategy is rendered ineffective as a result of the control/treatment identification strategy previously established in Propositions 1 and 2. That is, since the control group in our study is the first surgical case of each day in each operating room, there can *never* be a scenario in which a surgical case is coded to be a member of the control group *and* also have a surgeon on the immediately preceding case which possesses (or does not possess) prior process experience. By definition, the control group in our study does not have a case immediately preceding it, thus we are unable to code for the level of prior process experience on the surgeon in cases immediately preceding a control group case. Due to this empirical challenge, the DDD model is not a valid option for investigating the moderating effect of surgeon prior process experience. This empirical challenge, along with our identification strategy of the surgeon whose prior process experience moderates the relationship between the intraoperative IT prompt and holding room length of stay, also prevents us from splitting the control group by types of surgeons. However, based on the definition of control group cases (section 4 above), because the operating room is prepped and available to receive the first case, the first case of each day in each operating room will not be affected by any preceding, or in progress, surgical case or surgeon.

We resolve this empirical issue by employing a sample-splitting technique which has been recently utilized in studies that employ DID methodology (e.g., Janakiraman et al. 2018, Tucker et al. 2013). We note that this technique also aids in alleviating concerns of multicollinearity. This technique involves splitting the treatment group of surgical cases into two distinct subsamples. One treatment group subsample is composed of treatment group cases in which the surgeon on the immediately preceding surgical case (in the same operating room location) had prior process experience. The other treatment group subsample is composed of treatment group cases in which the surgeon on the immediately preceding surgical case (in the same operating room location) lacks prior process experience. We denote these two treatment subgroups as *Trtmnt_Prior* and *Trtmnt_NoPrior*, respectively. Upon appropriately dividing the treatment group, and following the procedures established by prior literature, we then proceed with estimating two separate DID models. In the first DID model, we interact the presence of the intraoperative prompt with treatment group cases which are immediately preceded by a surgical case (in the same operating room location) in which the preceding surgeon has extensive prior on-site process experience ($Prompt_t \times Trtmnt\_Prior_{ij}$). In the second DID model, we interact the presence of the intraoperative prompt with treatment group cases which are immediately preceded by a surgical case (in the same operating room location) in which the preceding surgeon lacks prior on-site process experience ($Prompt_t \times Trtmnt\_NoPrior_{ij}$).

Upon establishing the new DID econometric specification, we estimate Equations 3 and 4 to assess the moderating impact of a surgeon's prior process experience on the effectiveness of the intraoperative IT prompt. Table 4 (Column 1) displays results of the DID estimation when the prior surgeon has pre-intervention process experience. The coefficient associated with the interaction term, $Prompt_t \times Trtmnt\_Prior_{ij}$ captures the percentage reduction in treatment patients'

Table 4 Impact of Intraoperative IT Prompt: The Role of Surgeon Prior Process Experience

DV: ln(TiHR)	(1) Prior experience	(2) No prior experience
Prompt	0.028*** (0.0087)	0.027*** (0.0078)
Trtmnt_Prior	0.150*** (0.0159)	
Trtmnt_NoPrior		0.187*** (0.0187)
Prompt × Trtmnt_Prior	−0.118*** (0.0156)	
Prompt × Trtmnt_NoPrior		−0.083*** (0.0189)
OR Time (in mins)	3.59×10^{-04}*** (3.81×10^{-05})	2.64×10^{-04}*** (4.05×10^{-05})
Constant	4.593*** (0.0811)	4.626*** (0.0823)
Surgeon Fixed Effects	Yes	Yes
Patient Class Fixed Effects	Yes	Yes
Location Fixed Effects	Yes	Yes
Time Fixed Effects	Yes	Yes
Day Fixed Effects	Yes	Yes
Month Fixed Effects	Yes	Yes
Number of observations	17,558	12,309
R^2	0.212	0.239

Notes: Robust standard errors clustered by procedure type are in parentheses; ***$p < 0.01$.

holding room length of stay as a result of the implementation of the intraoperative IT prompt when the surgeon on the immediately preceding case (in the same operating room location) had pre-intervention process experience.

We find that patients undergoing a surgical procedure in the treatment group following the introduction of the IT intraoperative prompt, in which the prior surgeon had pre-intervention process experience, had a highly significant reduction in holding room length of stay ($\beta_3 = -0.118$; p-value = 0.000). More specifically, such patients experience an 11.8% reduction in holding room length of stay, on average (all other independent variables held constant). By comparison, the results in Table 4 (Column 2) show a less negative, though still significant ($\beta_3 = -0.083$; p-value = 0.000), reduction in holding room length of stay associated with patients undergoing a surgical procedure in the treatment group immediately following a case conducted by a surgeon without pre-intervention process experience. Once again, both models were tested relative to the pre-intervention period of 9,696 available surgical cases.

Although this result appears to indicate that surgeons with pre-intervention process experience yield greater reductions in holding room length of stay, we need to analytically examine whether the two treatment group subsamples produce DID coefficients which are significantly different from one another. Therefore, we conduct an equality test in which the null hypothesis states that the two coefficients, $Prompt_t \times Trtmnt\_Prior_{ij}$ and $Prompt_t \times Trtmnt\_NoPrior_{ij}$, are equal. Results of the test ($\chi^2 = 3.61$; p-value = 0.0575) reject the null hypothesis that the two DID coefficients are equal. Therefore, we conclude that a surgeon's prior on-site process experience amplifies the improvements to holding room length of stay previously demonstrated following the implementation of the intraoperative IT prompt, lending support for Proposition 3.

6. Robustness Checks and Supplementary Analyses

6.1. Validation for Impact of Intraoperative IT Prompt

To provide a more robust statistical assessment of the incremental effect of the intraoperative prompt, above and beyond the effects of the centralized decision-making process change, we have executed a DID robustness check directly comparing the 11,517 available surgical cases after the implementation of the intraoperative IT prompt with the 11,844 available surgical cases that were conducted after the centralized decision-making change (but before the introduction of the intraoperative IT prompt). These results are displayed in Table 5 and support the main results previously presented in Table 3.

The intraoperative IT prompt results in a highly significant reduction ($\beta_3 = -0.070$; p-value = 0.000) in the holding room length of stay for patients in the treatment group. This result indicates that patients in the treatment group that undergo a surgical procedure after the implementation of the intraoperative IT prompt change experience a 7.0% reduction in holding room length of stay, on average (all other independent variables held constant). We note that this result is in comparison to the 11,844 available surgical cases which occurred after the implementation of the centralized decision-making change (excluding the 9696 available surgical cases in the pre-intervention period). Combining the 7.0% reduction from the addition of the intraoperative prompt with the previously demonstrated 3.4% (results from test of Proposition 1 in Table 2) reduction following the implementation of the centralized decision-making change, the overall reduction in holding room time following both process changes is generally consistent with the reduction reported in Table 3 (which compared the 11,517 available cases after the intraoperative prompt with the 9696 available cases occurring before any changes were implemented).

6.2. Alternative Clustering

All of the DID models in our study include standard errors clustered by the CPT code associated with each surgical case. CPT codes represent a surgical procedure classification that was developed primarily for use in patient billing. To check the robustness of our clustering strategy, we also estimated models using an alternative clustering classification. Level codes are a grouping system which aggregates across multiple surgical procedures that have similar levels of complexity and require analogous levels of resources. Results for this robustness check indicate general consistency in magnitude, directionality, and level of statistical significance with those presented in our main analysis.[5] This finding confirms that clustering by an alternative surgical classification system does not diminish the impact of the process changes implemented in our study setting. Further empirical investigation of level codes supports our prior assertion that there is no systematic bias in the assignment of cases to the control and treatment groups, which if present, may lead to confounding of our empirical findings.

6.3. Support for Selection of Control Variables

Prior literature in management science and economics that employ the DID methodology have used fixed effects to account for unobserved effects (Forman et al. 2009, Goldfarb and Tucker 2011). Recall that our dataset is composed of prescheduled, elective cases, in which the surgeon on each case is scheduled prior to the patient's time in the holding room. As a result, we account for potential confounding effects stemming from individual surgeons by incorporating surgeon fixed effects. However, as an additional robustness check, we re-estimated the main models related to testing of Proposition 1 (impact of centralized decision-making) and Proposition 2 (impact of intraoperative IT prompt) *without* the inclusion of surgeon fixed effects. The results of these alternative models are substantively similar to the main results reported in Tables 2 and 3, increasing the validity of our main findings. We also conducted a similar re-estimation of our main findings without the intraoperative time control variable. Results remain robust to those reported in our main analysis and an equality test comparing DID coefficients in models with and without the intraoperative time control confirms that inclusion of this control does not improperly bias our findings.

6.4. Post Hoc Testing

With respect to the impact of our process changes on surgical quality, we carefully reviewed the surgical literature seeking evidence of a link between the amount of time patients spend in the holding room and the quality outcomes of their surgical procedures. To the best of our knowledge, no such studies exist establishing either a theoretical or empirical basis for a relationship between the time a patient spends in the holding room and any subsequent impact on the quality of the ensuing surgical procedure. Despite this current lack of theoretical evidence, we believe it prudent to investigate any potential changes to patient safety which may have occurred during the period following implementation of our process changes. To do so, we gathered and analyzed surgical quality ratings for our study hospital, comparing a period before any of our process changes were implemented against a period of time in the post-implementation phase of our study.

Leveraging healthcare quality data published by the Dartmouth Atlas of Healthcare,[6] we gathered surgical outcome data on three measures for our study hospital. These measures are as follows: percent of surgical patients readmitted within 30 days of initial discharge, percent of surgical patients seeing a primary care physician within 14 days of discharge to home, and percent of surgical patients having an emergency room visit within 30 days of initial discharge. To compare surgical quality outcomes before and after our interventions, we analyzed 12 months of surgical quality data at our study hospital (for each of the three measures) for a period of time that occurred immediately prior to our interventions, as compared to 12 months of surgical quality data that encompass the introduction of both of our process interventions. As added robustness, we gathered an additional 12 months of historical surgical quality outcome data which occurred well before the introduction of our process interventions, resulting in three consecutive years of surgical quality outcome data for each of the three measures.

To examine whether the three time periods have equal means, we conducted an analysis of variance (ANOVA) for each of the three surgical quality measures. The null hypothesis for each ANOVA test is that the means from the three time periods are equal ($\mu_1 = \mu_2 = \mu_3$). Testing equal means for the percent of surgical patients readmitted within 30 days of initial discharge (F-value = 2.727; p-value = 0.065), we fail to reject the null hypothesis and thus conclude that there are no differences in the means across all three time periods. Testing equal means for the percent of surgical patients seeing a primary care physician within 14 days of discharge to home (F-value = 0.113; p-value = 0.893), we fail to reject the null hypothesis and conclude that there are no differences in the means across all three time periods. Finally, testing equal means for the percent of surgical patients having an emergency room visit within 30 days of initial discharge (F-value = 0.463; p-value = 0.629), we fail to reject the null hypothesis and conclude that there are no differences in the means across all three time periods. Taken together, these findings, analyzed within our study hospital over time, provide empirical support that the process interventions in the focal hospital did not result in a negative impact on surgical outcome quality.

7. Discussion

Various types of communication have been previously implemented in the perioperative arena to improve clinical outcomes and reduce mortality and complications. Examples of these communication changes include preoperative "time-outs" and pre-closure supply checklists (Gawande 2009). Preoperative "time-outs" have proven effective in reducing death rates and complication rates while pre-closure supply checklists have been shown to reduce the number of retained foreign bodies following surgical procedures (Haynes et al. 2009, Stawicki et al. 2009). Given the clinical success achieved through focused communication and the evidence of benefits from centralized decision-making in the supply chain literature, our study builds upon both principles to demonstrate improvements to preoperative patient flow. A critical contrast between the aforementioned types of focused communication and our study is that the preoperative "time-out" and pre-closure supply checklists increase the number of employees involved in the decision-making process whereas the operational process change in our setting decreases the number of personnel involved in the decision-making process.

Within the healthcare operations literature, our study has some parallels to Song et al.'s (2015b) examination of queue management in the emergency department, in that both studies investigate the impact of operational changes on patient processing time in a healthcare service setting. Song et al., (2015b) target patient processing outcomes in the emergency department which are directly influenced by patient heterogeneity. In contrast, we have selected a dependent variable, preoperative processing time, in which patient heterogeneity is present but can be managed through a focus on targeting standards for patient arrival to and departure from the holding room. In other words, by isolating a component of the healthcare service process which can be improved regardless of patient differences, we are able to demonstrate efficiency gains related to the centralization of resources. Further, Song et al. (2015b) find evidence that strategic physician behavior (in dedicated queuing systems) generates greater service process efficiencies, in part due to increased levels of ownership by strategic servers. In contrast, our study extols the benefits of server actions when there is very little personal benefit to the server. Recall in our context that a surgeon's response to the intraoperative IT prompt generates efficiency benefits to the surgeon completing the next procedure scheduled for the room. That is, surgeons are behaving in a manner that is beneficial to their colleagues and overall departmental efficiency, not in a manner that is in their own strategic interest. Taken together, our study contributes to the healthcare operations literature by demonstrating the application of supply chain management principles to improve a distinct area of patient flow, in spite of the operational complications introduced by the presence of patient heterogeneity.

Perhaps more importantly, the results of our study provide evidence that an intraoperative IT prompt which enhances the amount, and accuracy, of shared information amplifies the benefits of centralized decision-making. Further building upon prior literature that presents evidence of the differential impact of surgeon experience on clinical performance outcomes (Neumayer et al. 2005, Sosa et al. 1998), our results provide support for a complementarity effect between surgeon prior process experience and operational process changes. With this result in mind, hospital administrators and surgical process owners should carefully consider the prior process experience of their surgical staff to optimize the benefits generated from an operational process redesign. A reduction in the amount of time that a patient spends in the holding room has significant implications for the utilization of operating room time. More efficient patient flow to the operating room reduces the amount of downtime present in one of the most highly profitable, and costly, resources within a hospital.

One of the primary monetary benefits of our process changes is that they were executed with no major costs to the hospital. In fact, the change to centralize decision-making was executed at effectively no cost to the perioperative services department or hospital. The employees who became responsible for calling the next scheduled patient to the holding room were already employed within the perioperative department. In lieu of hiring additional employees at a major annual salary expense, select existing employees received modified work responsibilities which aligned with the new role required to implement the centralized decision-making change.

With respect to the intraoperative IT prompt implementation, minimal IT development was required. Fortunately for our study hospital, at the time of implementation, they employed an electronic medical record system that had been developed and maintained by an in-house IT team. As a result, some software development was required; however, this cost was limited due to the availability of a hospital-employed software development team. The intraoperative IT prompt development process was expeditious due to the intricate knowledge of the software development team, many of whom had participated in the foundational build of the software in the past. As a result, the intraoperative IT prompt was developed in one week, at a labor cost of approximately $10,000. We acknowledge that the development cost and timeline would likely be different with an off-the-shelf electronic medical record system; however, we are unable to reliably estimate the cost of other provider software solutions to a hospital.

Another benefit of our process change solutions was the minimal staff training required. Since this was just a process workflow change, staff training did not require any major investments of labor training costs, such as classroom or simulation learning. Instead, training was primarily focused on staff communication, which took place in existing daily huddles and occurred on an ongoing basis for weeks before and after each process change. Because staff did not have to receive training outside of their normal working duties, there was very little training cost associated with the integration of our process changes into existing perioperative processes. Lastly, the efficiency benefits resulting from our process changes helped the hospital avoid the unnecessary and logistically challenging expense of constructing additional holding room space within the perioperative environment. We believe that these minor development and implementation costs, combined with the substantial cost savings of avoided construction, make our process changes an exceptionally valuable solution to enhance the efficiency of surgical holding rooms at little added cost.

In addition to the significant financial impact to the hospital, we also believe there may be an emotional and psychological benefit to each patient involved in the process (Tiwari et al. 2017). Awaiting a surgical procedure is a particularly stressful time during a patient's hospital stay (Calvin and Lane 1999, Gilmartin and Wright 2008). Although we were unable to empirically test the reduction in patient anxiety resulting from our process changes, future research may consider the benefits to patient satisfaction as a result of improvements to preoperative processing time. Such a benefit is seldom considered in the operations and manufacturing literature when studying production processes of consumer products yet is a critical consideration in the healthcare services context (Graham and Conley 1971, Johnston 1980).

While we have taken careful consideration to design our analysis for generalizability to perioperative departments of varying size and acuity, we acknowledge the limitations associated with applying our findings to perioperative areas of differing characteristics. Variable modes of surgical practice along with the intricacies of hospital practice and patient characteristics limit our ability to form conclusions about the effectiveness of our methods in other surgical settings. Although we feel strongly about the benefits of targeted modifications to improve preoperative flow, we cannot definitively conclude uniform applicability to all hospitals.

Although some evidence exists suggesting there is a considerable opportunity to integrate process flow management into existing perioperative IT systems (Bloomfield and Feinglass 2008), future research considerations should incorporate additional surgical cases from other hospitals to expand the generalizability of our study findings. Further work may also investigate specialty surgical hospitals to determine if the effects of process changes differ in focused settings. In our study context, the ability of the perioperative environment to fully realize the potential benefits of the intraoperative IT prompt is contingent upon centralized decision-making that aids sharing of knowledge between surgeons and the centralized decision-makers. Future studies may examine issues related to the sequencing of centralized decision-making and intraoperative IT prompts, along with various types of IT prompts on process outcomes. Although our study focuses on the role of surgeon process experience, future work may also investigate the relative level of combined experience between the surgeon conducting a case and the centralized decision-maker. Finally, future work may determine the impact of similar changes in other areas of high profitability within the healthcare industry, such as outpatient settings and emergency response services.

In summary, we conclude substantial process benefits resulting from the implementation of centralized decision-making and an intraoperative IT prompt in the perioperative services environment. Our study contributes to the operations management and healthcare literature by providing empirical evidence of the impact of centralized decision-making

Table 5 Impact of Intraoperative IT Prompt (Relative to the Impact of Centralized Decision-making)

	DV: ln(TiHR)
Prompt	−0.020 (0.0148)
Trtmnt	0.118*** (0.0129)
Prompt × Trtmnt	−0.070*** (0.0108)
OR Time (in mins)	3.89×10^{-04}*** (4.03×10^{-05})
Constant	4.500*** (0.0627)
Surgeon Fixed Effects	Yes
Patient Class Fixed Effects	Yes
Location Fixed Effects	Yes
Time Fixed Effects	Yes
Day Fixed Effects	Yes
Month Fixed Effects	Yes
Number of observations	23,359
R^2	0.210

Notes: Robust standard errors clustered by procedure type are in parentheses; ***$p < 0.01$.

and IT to reduce the barriers of information flow, resulting in significant improvements in preoperative patient flow. We also provide evidence of the beneficial use of shared health information to drive operational improvements, above and beyond the well-publicized benefits to quality of clinical care.

Notes

[1]For a full review of the benefits and consequences of the various types of human-computer interruptions, we refer readers to McFarlane (2002).

[2]Timing of the prompt was collaboratively selected by surgeons and anesthesiologists in our focal hospital as the appropriate time to receive the intraoperative prompt because they were confident that, 90 minutes (60 minutes from the end of the previous surgery plus 30 minutes to clean the operating room) was ample time for preoperative processing, regardless of the type of procedure for which the patient was about to undergo.

[3]Empirical analysis of surgical cases excluded due to missing data indicates no evidence of systematic bias. More information is available from the authors upon request.

[4]Empirical analysis of the relative acuity and resource utilization of cases assigned to the control and treatment groups indicates no evidence of systematic bias.

[5]In the interest of space, we do not provide results of the alternative models. More information is available from the authors upon request.

[6]The data set forth at Hospital Discharges and Post-Acute Care of publication/press release were obtained from Dartmouth Atlas Data website, which was funded by the Robert Wood Johnson Foundation, The Dartmouth Clinical and Translational Science Institute, under award number UL1TR001086 from the National Center for Advancing Translational Sciences (NCATS) of the National Institutes of Health (NIH), and in part, by the National Institute of Aging, under award number U01 AG046830.

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